

CHIMBAY, Uzbekistan

WMO: 382620

Lat: 42.950N

Lon: 59.800E

Elev: 305

StdP: 14.53

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.0	2.7	-10.4	3.2	-1.7	-5.5	4.1	3.7	20.3	28.8	18.2	28.1	5.6	50	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.6	101.6	72.1	98.7	70.7	96.0	69.5	75.5	95.5	73.5	93.4	71.7	92.1	4.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	109.0	87.4	66.4	98.1	85.5	64.0	90.3	84.1	39.2	95.2	37.2	93.5	35.6	92.0	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.0	15.8	-5.5	107.4	7.5	3.3	-10.9	109.8	-15.3	111.7	-19.6	113.5	-25.0	115.9		
(5)			-5.6	78.9	6.8	3.2	-10.5	81.2	-14.5	83.1	-18.3	85.0	-23.3	87.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.6	24.3	28.6	43.3	58.3	71.0	79.4	82.5	79.0	66.7	53.1	38.6	28.3		
	DBStd	22.25	10.57	10.92	10.04	9.26	6.84	5.61	4.93	5.94	7.05	8.47	9.75	9.23		
	HDD50	2779	798	600	254	30	0	0	0	0	1	66	357	673		
	HDD65	5590	1263	1019	674	239	26	1	0	1	61	376	792	1138		
	CDD50	4445	0	1	46	279	651	882	1006	900	503	162	15	0		
	CDD65	1781	0	0	1	38	212	433	541	436	113	7	0	0		
	CDH74	22988	0	0	33	517	2558	5736	7280	5374	1335	150	5	0		
	CDH80	12229	0	0	7	176	1124	3196	4232	2969	499	25	1	0		
	WSAvg	5.7	5.7	6.4	7.2	6.9	6.6	6.0	5.1	4.6	4.8	4.8	5.0	5.3		
	PrecAvg	4.8	0.5	0.4	0.7	0.7	0.7	0.2	0.1	0.1	0.1	0.4	0.4	0.5		
(15) Precipitation	PrecMax	10.0	1.6	2.1	2.2	3.3	2.8	1.2	1.0	0.6	0.8	1.5	1.8	2.0		
	PrecMin	2.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.8	0.3	0.5	0.6	0.7	0.7	0.2	0.2	0.2	0.2	0.3	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	61.1	78.7	91.2	97.4	103.7	106.6	104.4	95.7	83.3	69.7	54.1		
		MCWB	43.6	49.8	57.4	65.3	67.2	73.8	73.0	74.0	69.4	60.5	54.7	46.1		
	2%	DB	44.9	52.5	70.0	85.4	93.1	99.9	102.3	99.9	89.8	78.1	62.4	47.9		
		MCWB	39.6	44.0	54.3	62.4	66.9	71.1	72.0	72.0	66.5	58.9	51.1	41.2		
	5%	DB	40.3	46.9	64.5	80.1	90.0	97.0	99.0	96.8	85.7	73.8	57.6	43.7		
		MCWB	36.4	40.2	50.8	59.2	65.5	69.5	70.7	70.5	64.0	56.8	47.9	38.4		
	10%	DB	37.1	42.4	59.2	75.0	86.6	94.1	96.1	93.6	81.7	69.0	52.9	40.3		
		MCWB	33.8	37.0	48.1	56.8	63.8	68.0	69.8	69.1	62.1	54.2	45.4	35.8		
	0.4%	WB	44.2	50.5	58.5	67.4	73.2	77.9	77.7	79.9	71.4	63.1	56.8	46.8		
		MCDB	50.9	60.5	76.4	86.3	91.9	96.4	97.2	98.5	89.6	79.4	66.5	52.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.8	44.6	54.8	64.0	69.2	74.5	74.9	75.2	68.2	60.0	52.4	41.9		
		MCDB	44.1	51.2	68.3	82.0	88.2	95.1	95.7	94.4	88.0	75.5	61.1	47.1		
	5%	WB	36.7	40.8	51.7	61.0	66.9	72.1	73.2	72.6	65.6	57.4	48.9	38.7		
		MCDB	40.1	46.5	63.4	77.6	85.6	92.7	93.5	92.4	82.6	71.7	56.5	43.2		
	10%	WB	34.1	37.4	48.6	58.1	65.1	69.9	71.7	70.3	63.1	54.9	45.6	36.1		
		MCDB	37.0	42.1	58.3	72.6	83.2	89.6	92.4	90.4	79.1	67.8	52.4	40.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	17.5	20.9	23.0	24.4	24.9	24.6	26.0	26.3	25.0	18.8	14.8		
		MCDBR	17.6	23.7	28.9	29.8	29.5	29.4	29.0	29.8	30.3	31.3	26.0	19.5		
		MCWBR	13.9	16.8	16.5	13.5	12.5	11.5	10.9	12.3	14.2	16.7	16.5	15.0		
		MCDBR	17.0	22.7	26.5	28.3	26.4	27.5	26.8	28.7	29.0	28.4	23.0	17.9		
	5% WB	MCWBR	13.7	16.5	15.8	14.4	12.9	12.4	11.3	13.0	14.7	16.3	15.4	14.2		
(40) Clear-Sky Solar Irradiance	taub		0.377	0.405	0.467	0.522	0.526	0.511	0.493	0.469	0.441	0.446	0.410	0.381		
	taud		2.061	2.007	1.914	1.821	1.858	1.927	2.001	2.035	2.050	1.983	2.036	2.058		
	Ebn at Noon		224	240	241	238	242	246	249	250	246	223	209	208		
	Edh at Noon		35	43	54	64	63	59	55	51	47	44	36	32		
(44) All-Sky Solar Radiation	RadAvg		479	797	1241	1699	2108	2364	2278	2087	1645	1086	648	442		
	RadStd		56	96	105	120	120	103	78	68	67	62	77	59		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	+1.53	N/A	N/A	N/A	N/A	N/A	+95
	N/A	N/A	N/A	+1.49	N/A	-1.21	N/A	N/A	N/A	+94

Nomenclature: See separate page